Section 6

Section 1.5 · 2031 Concept Specifications

PhysicsWallah · 10 innovation ideas (2 per STEEP sector), each backtracked from a real 2026 artifact in our Stage 4 workflows · 2026 → 2031

👥 SOCIAL · IDEA 01

Verified Teacher Handover Live

🔭 2031 SYSTEM ROLE

When a teacher leaves, the outgoing and incoming teacher run a live handover class for the batch inside the app. Students hear the reason first-hand and get a clear plan for the rest of the year.

🧱 2026 BASELINE FRICTION RESOLVED

⚡ Rumour screenshots forwarded in Telegram / WhatsApp batch groups, and PW's late in-app notice (Workflow 1).

🤝 CULTURAL ERGONOMICS & INTERACTION MODEL

A Hinglish live class in the familiar "bhaiya / didi" style, with both teachers on screen. Students ask questions in chat; the recording is pinned for anyone who missed it.

🔐 CONTEXTUAL INTEGRITY AUDIT

Sender → Recipient: PW app (both teachers) → enrolled batch students
Subject · Info type: The batch · faculty-change reason, new timetable, transition plan
Transmission principle: Shared only inside the enrolled batch; the leaving teacher agrees to what is said about their exit.

⚡ TECHNOLOGICAL · IDEA 03

Portable Learning Passport

🔭 2031 SYSTEM ROLE

The student's verified learning record (scores, chapters done, doubt summary) lives in an APAAR-linked wallet and moves with them across batches and platforms. PW reads it with consent instead of asking again.

🧱 2026 BASELINE FRICTION RESOLVED

🟡 Re-confirming batch and enrollment data PW already holds (Workflow 2).

🤝 CULTURAL ERGONOMICS & INTERACTION MODEL

One "Share my progress" tap, protected by a PIN on the shared family phone. Shown as a simple chapter map, not a data dump.

🔐 CONTEXTUAL INTEGRITY AUDIT

Sender → Recipient: Student's learning wallet (DigiLocker / APAAR) → PW batch system or another platform
Subject · Info type: The student · verified progress summary
Transmission principle: Student-controlled, purpose-limited consent token that expires when the batch ends.

🌿 ENVIRONMENTAL · IDEA 05

Offline Classroom Box

🔭 2031 SYSTEM ROLE

A small device at each offline centre caches lectures, marks attendance by QR or student card, and syncs with PW when the internet returns. Paper registers and end-of-week retyping disappear.

🧱 2026 BASELINE FRICTION RESOLVED

🛠️ Paper attendance register retyped by hand: the 🔴 manual media break (Workflow 3).

🤝 CULTURAL ERGONOMICS & INTERACTION MODEL

Works with no internet. Students tap their ID card at the door like a metro gate; the centre manager sees a simple daily list in the local language.

🔐 CONTEXTUAL INTEGRITY AUDIT

Sender → Recipient: Centre box → PW central system
Subject · Info type: Students at the centre · attendance time-stamps
Transmission principle: Collected only for attendance and safety; synced encrypted; each parent sees only their own child's record.

💰 ECONOMIC · IDEA 07

Teacher-Linked Fair Refund

🔭 2031 SYSTEM ROLE

Fees are paid month by month and tied to the promised teacher. If that teacher leaves, the unused amount is refunded automatically, with no ticket needed.

🧱 2026 BASELINE FRICTION RESOLVED

⚡ Refund request ticket + 🛠️ fee receipt screenshot + ⏳ manual review queue (Workflow 4).

🤝 CULTURAL ERGONOMICS & INTERACTION MODEL

The parent gets a Hindi voice note: "Sir has left. ₹1,200 is coming back to your UPI today." One tap: stay with the new teacher, or keep the refund.

🔐 CONTEXTUAL INTEGRITY AUDIT

Sender → Recipient: PW billing system → parent's UPI app
Subject · Info type: The parent's fee · refund amount and reason
Transmission principle: Fee terms the parent accepted at purchase; no new data or documents asked for.

🏛️ POLITICAL / REGULATORY · IDEA 09

One-Tap Consent & Delete

🔭 2031 SYSTEM ROLE

Parents manage what PW can do with their child's data through a DPDP consent manager, with time-limited, purpose-tagged consent. Withdrawing consent or deleting data takes one tap, not emails and ID photos.

🧱 2026 BASELINE FRICTION RESOLVED

🛠️ ID photo sent again + ⚡ complaint email + ⏳ grievance-officer wait (Workflow 5).

🤝 CULTURAL ERGONOMICS & INTERACTION MODEL

Plain-language consent cards with icons and a Hindi voice explainer. Shows "who used your child's data this month".

🔐 CONTEXTUAL INTEGRITY AUDIT

Sender → Recipient: Parent's consent manager → PW data system
Subject · Info type: The child · consent status and deletion request
Transmission principle: DPDP Act consent and erasure rights, verified through the existing DigiLocker login (no new ID upload).

👥 SOCIAL · IDEA 02

Family Decision Room

🔭 2031 SYSTEM ROLE

A shared space where the student and parent see progress together and decide on renewal as a family. It replaces one-way reminders with a joint decision.

🧱 2026 BASELINE FRICTION RESOLVED

⚡ CRM renewal reminder that gets ignored; the parent decides without seeing the student's progress (Workflow 1).

🤝 CULTURAL ERGONOMICS & INTERACTION MODEL

One shared family view on one phone. The parent gets a Hindi voice summary; the student adds a voice note on why they want to stay or switch. Renewal needs both taps.

🔐 CONTEXTUAL INTEGRITY AUDIT

Sender → Recipient: PW app → student + linked parent
Subject · Info type: The student · attendance, test trend, teacher-continuity status (no private doubt chats)
Transmission principle: Parent consent for minors under DPDP; the student chooses which progress items the parent sees.

⚡ TECHNOLOGICAL · IDEA 04

Continuity AI Tutor

🔭 2031 SYSTEM ROLE

An AI tutor trained on the outgoing teacher's own lectures and notes covers the gap during a faculty swap. It keeps the same explanations and pace until the new teacher settles in.

🧱 2026 BASELINE FRICTION RESOLVED

⏳ Class gap while the new teacher onboards, and a bare "new faculty assigned" push alert (Workflow 2).

🤝 CULTURAL ERGONOMICS & INTERACTION MODEL

Same Hinglish style and worked examples, clearly labelled "AI, based on Sir's lectures". Students ask doubts by voice.

🔐 CONTEXTUAL INTEGRITY AUDIT

Sender → Recipient: PW content system → AI tutor → student
Subject · Info type: The teacher's lecture content · the student's doubts
Transmission principle: Teacher's contract licenses lectures for AI use with credit and royalty; doubts are used only to answer them.

🌿 ENVIRONMENTAL · IDEA 06

Voice & IVR Updates

🔭 2031 SYSTEM ROLE

Families without smartphones or with weak data get timetable and teacher-change updates as a voice call in their language. They press a key to confirm or ask for a callback.

🧱 2026 BASELINE FRICTION RESOLVED

🛠️ Printed timetable on the centre notice board that students miss (Workflow 3).

🤝 CULTURAL ERGONOMICS & INTERACTION MODEL

Hindi, Bhojpuri or Marathi voice on any basic phone. Give a missed call to hear it again; no reading needed.

🔐 CONTEXTUAL INTEGRITY AUDIT

Sender → Recipient: PW notification service → parent's / student's phone number
Subject · Info type: The batch · timetable and teacher change
Transmission principle: Only to numbers registered at admission; opt out any time; no marketing in the same call.

💰 ECONOMIC · IDEA 08

Trust Escrow

🔭 2031 SYSTEM ROLE

Part of the yearly fee sits in escrow and is released to PW month by month only while the promised teacher and timetable hold. Families feel safe paying upfront, and PW still gets steady income.

🧱 2026 BASELINE FRICTION RESOLVED

⚡ One-shot annual UPI autopay mandate, and the fight to claw money back (Workflow 4).

🤝 CULTURAL ERGONOMICS & INTERACTION MODEL

A simple "money safe" meter in the app shows how much is still protected, explained in Hindi like a familiar family savings kitty.

🔐 CONTEXTUAL INTEGRITY AUDIT

Sender → Recipient: Parent's bank → regulated escrow account → PW
Subject · Info type: The parent's payment · release triggers (teacher continuity, classes held)
Transmission principle: RBI-regulated escrow rules; release conditions written into the fee agreement.

🏛️ POLITICAL / REGULATORY · IDEA 10

Faculty Change Ledger

🔭 2031 SYSTEM ROLE

A regulator-visible ledger records every teacher promise, change and refund for each batch. Consumer authorities audit it automatically instead of waiting for complaints.

🧱 2026 BASELINE FRICTION RESOLVED

⚡ National Consumer Helpline complaint form, and "star teacher" ads that overpromise (Workflow 5).

🤝 CULTURAL ERGONOMICS & INTERACTION MODEL

Before buying, parents see a public "batch promise card" (teacher, dates, refund rule). A complaint becomes one tap that points to the ledger entry.

🔐 CONTEXTUAL INTEGRITY AUDIT

Sender → Recipient: PW → regulator audit node (e.g. CCPA)
Subject · Info type: Batches and faculty · change dates, refunds issued (no student personal data)
Transmission principle: Statutory audit mandate; aggregated data only, no individual student records.

Columns = STEEP sector · Each card names the exact 2026 artifact (🛠️ material / ⚡ digital) it removes · 🔴 media break · 🟡 redundant request · ⏳ latency gate